



Fair uplink bandwidth allocation and latency guarantee for mobile WiMAX using fuzzy adaptive deficit round robin

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ABSTRACT

The explosive demands of rich media applications with their diverse quality of service (QoS) requirements have continuously fuelled the needs for ever more powerful networks. One example of such a network is called WiMAX which is driven by WiMAX Forum based on IEEE 802.16 Wireless MAN standard. One of the issues that still remain open in WiMAX is the scheduling algorithm that goes to meet the QoS requirements. However, QoS provisioning of real-time and non real-time applications are frequently unstable due to insufficient allocation of bandwidth, which leads to degradation in latency guarantee and deterioration of overall system utilization. In this paper, an efficient bandwidth allocation algorithm for the uplink traffic in mobile WiMAX is proposed. Using intelligent systems approach upon the traffic service class information served by the base station (BS), an adaptive deadline-based scheme is designed. The scheme is fully dynamic to guarantee a specific maximum latency for real-time applications, besides improving fairness and throughput, giving due considerations to non real-time applications. The algorithm uses fuzzy logic control which is embedded in the scheduler; its function is to control and dynamically update the bandwidth required by the various service classes according to their respective priorities, maximum latency and throughput. Simulation results show that the proposed algorithm manages to optimize the overall system utilization while at the same time guarantee the maximum latency requirements for real-time traffic.

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1. Introduction

The growing popularity of wireless broadband services plus the explosive demands for a diversity of real-time applications such as voice-over-IP (VoIP), video streaming and gaming, have become key driving factors for the deployment of seamless and ubiquitous wireless access networks. The IEEE 802.16 standard for example, defines a broadband wireless access network for metropolitan area, also commercially known as WiMAX (IEEE, 2005, 2006). WiMAX was developed to meet the anticipated growth in the worldwide market for high bandwidth and real-time applications. However, IEEE 802.16 standard does not specify a bandwidth allocation algorithm to guarantee QoS, this is purposely done in order to allow service providers and vendors to innovate in this area and distinguish their products.

In wireless broadband access networks, the channels have to transport a wide variety of multimedia applications and this becomes challenging for network service providers to meet;

this is due to the scarce wireless resources available to satisfy all traffic demands with diverse QoS requirements. In WiMAX, the bandwidth resource management is divided into downlink (DL) and uplink (UL) direction controlled by a BS. The service classes' parameters can be differentiated and prioritized in these traffic directions. However, real-time applications such as video conferencing and gaming require a higher bandwidth allocation, which increases delay and reduces efficiency of the overall system. Traffic scheduling and bandwidth management schemes are two key mechanisms in WiMAX that are used to support the required QoS. A number of scheduling algorithms have been proposed to deal with the QoS requirements for the various service class' applications in WiMAX, but it is still necessary to develop appropriate bandwidth allocation schemes to guarantee satisfactory QoS.

In this paper, a fair and efficient bandwidth allocation algorithm for mobile WiMAX in the uplink direction is presented. Towards that end, a fuzzy logic system is developed as an embedded system for a new deadline-aware bandwidth allocation. This is referred here as fuzzy based adaptive deficit round robin scheduling or FADRR. The deadline is controlled and computed based on the input variables, namely maximum latency for real-time traffic and throughput for the non real-time traffic of all service classes. Therefore, the objective of FADRR is to reduce

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negative impact on the QoS metrics and ensure optimal bandwidth for all service flows, and at the same time to maintain fairness and conserve system resources. To the best of our knowledge, this is the earliest application of embedded intelligent mechanism to manipulate the queue behaviour and grant optimal bandwidth to real-time traffic queues. This algorithm can dynamically distinguish the data of each service type queue; and it adaptively allocates adequate bandwidth for real-time service types, considering their deadline, and improves access for non real-time connections. The performance of the algorithm has been compared by means of extensive simulation scenarios against some representative schemes proposed in this area, namely modified deficit round robin (MDRR) (Cisco), and customized deficit round robin (CDRR) (Laias and Awan, 2010). The results indicate that this scheme manages to provide improvements over both MDRR and CDRR in terms of delay, jitter, throughput and fairness.

2. IEEE 802.16 concept and overview

IEEE 802.16 is a wireless broadband standard that offers higher data rate over a metropolitan area of up to 70 km (IEEE, 2010). The standard identifies the frequency band, medium access control (MAC) and physical (PHY) layers for wireless broadband access. It uses orthogonal frequency division multiplexing access (OFDMA) as the multiplexing technology that distributes the bandwidth into numerous frequency sub-carriers. It manipulates the frequency range of the channel by the modulation code and modulate the information to the sub-carriers before transmission (Bacioccola et al., 2010). The available resources in OFDMA are known in the time domain (TD) as symbols and in the frequency domain (FD) as sub-carriers, which integrate into a sub-channel system. OFDM is used to provide the multiplexing of user's data streams on both uplink and downlink transmissions whose scope is beyond the scope of this paper.

To support QoS in wireless broadband networks, the MAC layer has to size up the downlink and uplink traffic. The received flows at the MAC layer are classified and related with the corresponding service class. Five service classes have been defined in IEEE 802.16. They are unsolicited grant service (UGS), extended real-time polling service (ertPS) and real-time polling service (rtPS), which known as real-time applications, while non real-time polling service (nrtPS) and best effort (BE) services support non real-time applications. The bandwidth allocation algorithm must ensure the QoS for different traffic types real-time and non real-time, at the same time utilizing efficiently the available bandwidth. While the DL scheduler in a BS easily allocate DL data to subscriber stations (SSs), the uplink traffic transmission relies on request and grant mechanism where the SSs can use the polling mechanism BW-REQ messages periodically to connect to the BS. Throughout the scheduling algorithm, the BS reserves the required bandwidth based on the number of slots required for each request in the uplink sub-frame. Along with all service classes, excluding the UGS and ertPS which are granted with a fixed bandwidth, rtPS must be taken into account due to the nature of variable packet sizes and also must guarantee the maximum-latency to satisfy the QoS requirement. The available bandwidth in the system is allocated to SSs based on the scheduling mechanism; this has the impact of satisfying the diverse QoS requirements while at the same time increase bandwidth utilization to improve the system capacity. The bandwidth allocation algorithms performed by the BS provides more significant end user satisfaction, therefore these algorithms must be designed and used cautiously to optimize the system utilization.

3. Related works

Scheduling and bandwidth allocation algorithms are two of the critical mechanisms to provide the required QoS in a packet

network. A number of bandwidth allocation algorithms have been studied to overcome the drawbacks that affect service class applications in WiMAX. A comprehensive survey and taxonomy of scheduling in IEEE 802.16e WiMAX networks can be found in Msadaa et al. (2010). This section provides overview of some key scheduling algorithms that have been proposed for WiMAX networks.

In Shreedhar and Varghese (1996), an amendment to both round robin (RR) and weighted round robin (WRR) which is known as deficit round robin (DRR) was proposed, where the scheduling process requires only $O(1)$ complexity to process a packet in the queue in addition to the implementation simplicity, at hardware with a reasonably low cost. However, real-time packets will have to wait until the scheduler completes serving other queues in this round including BE packets, the effect of which is that real-time packets may miss their deadline.

The method in Wongthavarawat and Ganz (2003) proposed a two-tier hierarchical algorithm. In the first-tier deficit fair priority queuing (DFPQ) allocates the total available bandwidth for DL and UL traffic, while at the second tier, it handles different queues by different conventional algorithms. The drawback is that DFPQ does not guarantee QoS for real-time services.

A scheduling algorithm for the rtPS was proposed in Ball et al. (2005). This algorithm manipulates a scheduling list that contains all the SSs that can be served at the next frame. However, the algorithm specifies that the SSs that has low transmission quality is suspended temporarily from the transmission list for a period of time. This mechanism is repeated periodically for all SSs. If the transmission quality is still low, the scheduler grants another suspended period of time.

In Ball et al. (2006), RR scheduler was investigated, whereby it allocates the available resources to all the SSs in a round-robin fashion without giving due priority to real-time applications. This algorithm is uncomplicated and simply manipulates the available resources among the SSs. Therefore it is considered inappropriate for traffic with diverse characteristics and QoS requirements. In order to distinguish the required bandwidth for each queue the authors in Cicconetti et al. (2006) handled the traffic in two ways; first, for traffic that does not require QoS, and next for traffic that do require QoS. However, fairness in this case was not considered. In Rath et al. (2006), the opportunistic deficit round robin (O-DRR) scheduler proposed an analytical method for getting an optimal polling interval for uplink data flow via the polling interval mechanism, the BS polls service flow periodically to make sure that the traffic delays are achieved. The system considering several situations, for instance, the SSs must ensure that the queue should not be empty as well as the receive SNR must exceed the threshold value. However, the allocation mechanism of the O-DRR algorithm leads to an additional overhead at the BS because it requires the manipulation of quantum size and a deficit count for each SSs, repeatedly. A modification to DRR known as modified deficit round robin (MDRR) was proposed in Cisco. It implements a quantum ϕ and a deficit counter (DC) for each service type queue. In each round the scheduler assigns the service type queue by DC value that is added by the value in every round; the scheduler transmits the packets till the DC empties or when the packet queue length is greater than the DC, and then move to the next queue. However, real-time packets will experience severe delay when the traffic in the system is heavy.

A latency and modulation aware bandwidth allocation algorithm called highest urgency first (HUF) was proposed in Lin et al. (2009). While HUF translates the data bytes into slots to investigate the effect from various adaptive modulation and coding scheme (MCS), it does not take full benefit from the MCS variations. Additionally, with HUF the request is discarded whenever its deadline is less than a frame duration, which is treated as a violation of maximum latency requirement.

A customized deficit round robin (CDRR) was proposed in [Laia and Awan \(2010\)](#). The algorithm takes care of real-time flow by adding a new queue to schedule real-time applications just prior to the deadline. However, this extra queue increases the delay for non real-time applications such as nrtPS and BE. The interruption caused by the transmission of non real-time packets in the extra queue will degrade the overall system throughput as well as violate the real-time application deadline for the packets in the rtPS queues. This is due to the interception of the extra queue for the real-time signal, which leads to increased overhead for the system which is not desirable, in particular, when the traffic is high. Moreover, assigning fixed weight to the queues lead to unfairness in resource sharing among non real-time applications.

In addition to the algorithms described above, there are other scheduling algorithms adopting different approaches which are as described in [Alsahag et al. \(2011\)](#), [Kao and Chuang \(2012\)](#), [Lin et al. \(2010\)](#). However, to date all bandwidth allocation algorithms are not capable to avoid violation of the deadline for real-time applications. The bandwidth allocation algorithm proposed in this paper differs from the previous works whereby it embeds an intelligent system that calculate and allocate adequate bandwidth dynamically for various types of traffic.

4. Fuzzy based adaptive deficit round robin uplink scheduler

By performing adaptive and dynamic scheduling methods based on traffic requirements, we can satisfy the demands better than in static methods. In this paper, we describe a new scheduler called FADRR. This is based on DRR with low complexity ([Shreedhar and Varghese, 1996](#)). FADRR uses expert systems based on fuzzy logic to adjust the service queue weights for real-time and non real-time applications. This can be implemented with an embedded system. The optimal bandwidth is granted dynamically for each service class. This enables another service type to experience a higher priority when the real-time connections have not yet reached their maximum latencies. Our scheme dynamically updates the weight of which is the amount of bandwidth assigned to the service type queue to determine the number of allowed packets to be transmitted in every round. This weight considers as a priority of the service type queue which represents the bandwidth allocated for each application request by the embedded fuzzy system output

weight ratio to provide the optimal bandwidth to each service class queue. The BS allocates the required bandwidth for the connection requests and updates an active list (L) of application requests at the start of every scheduling round. Our goal is to ensure that we choose an optimal bandwidth granted for a service flow requests in each queue such that every application request is served within an allowed period, while still being fair to the different service classes. The components and operations of the FADRR algorithm are illustrated in [Fig. 1](#).

4.1. Optimal bandwidth for service type queue

In this work, the scheduler in the BS prioritizes the SSs to determine the bandwidth and traffic requirements for each queue. It is important to allocate adequate bandwidth to the diverse SSs demands, in which the efficiency and fairness are maintained. The scheduling algorithm in this work satisfies fair bandwidth allocation in addition to guaranteeing maximum latency for real-time applications. The transmission unit in IEEE 802.16 is time slot based, and the frame duration is divided into a fixed number of time slots, which may vary from one frame to another depending on the adaptive modulation coding (AMC) mode used in response to the channel condition. Therefore, firstly we divide the available bandwidth in the uplink sub-frames into time slots because a slot in Mobile WiMAX PHY ([Lin et al., 2009](#)) contains 48 data sub-carriers. The slot capacity (S_c) is calculated as follows:

$$S_c = \frac{48 \times mod_{bits} \times coding_{rate}}{8} \quad (1)$$

where, mod_{bits} is the number of bits comprises the symbols in a modulation scheme, and $coding_{rate}$ is the coding rate for the modulation scheme.

The number of frames F_n can be calculated based on the equation

$$F_n = \frac{R_{Max}}{FD} \quad (2)$$

where, R_{Max} is the maximum rate in the system and FD is the frame duration. When a new request application starts, it must firstly be translated into a number of slots as

$$S_i = \frac{Req_i}{S_c} \quad (3)$$

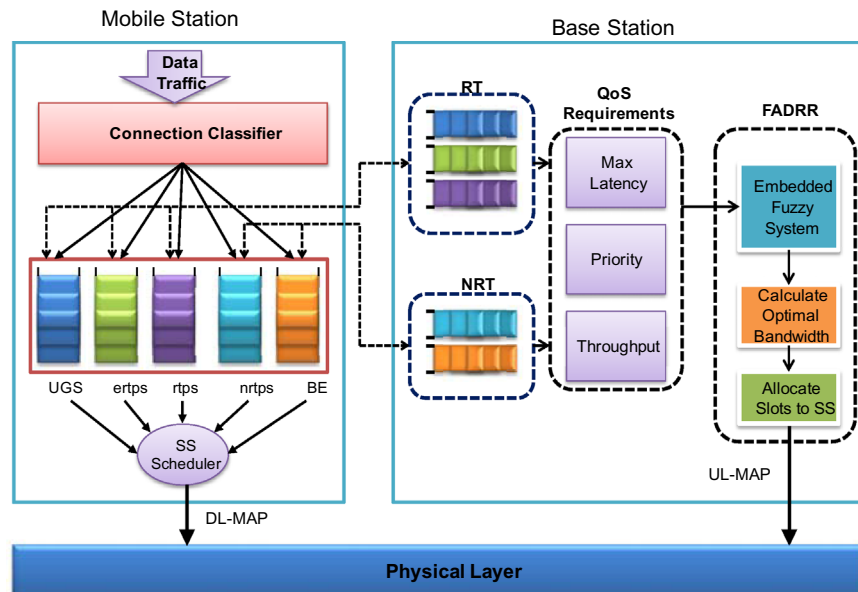


Fig. 1. Schematic of the proposed scheduler design with FADRR.

where, Req_i denotes the requested size and S_i represents the required number of slots for one request. In the service type queues, let M_i be the maximum packet length in Q_i , in Shreedhar and Varghese (1996) it has been evaluated that the following formula must be achieved from DRR in order to demonstrate O(1) complexity:

$$\forall i, \phi \geq M_i \quad (4)$$

This means that each of the queue's quantum ϕ must be large enough to contain the maximum packet length for that flow. Assume that FADRR algorithm is used to schedule packets from N queues with a maximum rate R_{Max} . The frame length is given as

$$F = \sum_{i=1}^N \phi_i \quad (5)$$

In FADRR the minimum guaranteed rate of Q_i is

$$R_{min} = \frac{\phi_i}{F} \times R_{Max} \quad (6)$$

As for maximum latency due to the embedded fuzzy system we have to consider the scheduling latency which is set to be equal to the maximum duration that ahead-of-line (HoL) packet can be delayed due to scheduling decisions and the calculation delay to compute the quantum values for each queue. For DRR, a tight delay bound for this metric has been computed in Lenzini et al. (2004). The scheduling delay decision of FADRR can be calculated as

$$SL_i = \frac{1}{R_{Max}} \left[F - \phi_i \times \left\lceil \frac{M_i}{\phi_i} \right\rceil + \sum_{i=1}^N M_i \right] \quad (7)$$

The weight required for all connections in the queue is calculated as follows: j is the number of subscribers, i the required weight for this subscriber.

$$W = \sum_{j=1}^N w_i \quad (8)$$

where w_i indicates the priority of service type queue can be achieved by the following equation:

$$w_i = \frac{R_{min(i)}}{\sum_{i=0}^N R_{min(i)}} \quad (9)$$

where $R_{min(i)}$ is the minimum rate for the application request i assigned in WiMAX based on the priority of the service class, and N is the total number of application requests. Regarding the service classes such as UGS, ertPS and rtPS, the maximum latency parameter is expected to be guaranteed for real-time applications. Thus, in this algorithm, this is defined as deadline (D) given as

$$D = \frac{ML + SL_i}{FD} \quad (10)$$

where ML represents the maximum latency of the service flow (SF), SL_i is the scheduler delay for selecting HoL packet and FD represents the frame duration. To ensure adequate bandwidth, we have to calculate the amount of bandwidth required for various service classes. Therefore, in this paper we develop an embedded fuzzy system to dynamically compute the required bandwidth more accurately and with low complexity.

The embedded fuzzy system works by selecting two input variables. First is maximum latency (ML) of the HOL packet for the corresponding queues such as UGS, ertPS and rtPS class services, denoted as RT_{ML} . The second variable is throughput for non real-time class services such as nrtPS and BE denoted by NRT_{Thru} . Thus, the systems state vector input variables can be defined as

$$\chi = (RT_{ML}, NRT_{Thru}) \quad (11)$$

In the proposed algorithm, each of the queue is assigned a value given by ϕ which measure the amount of packets that must be transmitted in a service round. In order to satisfy the QoS

requirements and maximize system utilization, we calculate the amount of bandwidth required for each service type queue's ϕ by considering the normalized ratio resulting from the embedded fuzzy logic system for each application request in the queue and the traffic status in the overall system considering the QoS requirements specifically the deadline of a real-time application and minimum reserved rate for overall requests. The weight ratio is calculated from the embedded fuzzy method executed based on the maximum latency for each real-time request and throughput for non real-time request, this weight provides for each service class queue, participated as a ratio between real-time and non real-time traffic, which enables the scheduler to make a decision based on the QoS and network constraints, in order to allocate the bandwidth for each service class queue. We compute the quantum value ϕ as the following equation:

$$\phi_i = \gamma \times \left(\frac{S_i \times \left(\frac{w_i}{W} \right)}{D} \right) + M_i \quad (12)$$

where γ is the an optimal bandwidth ratio obtained from the embedded fuzzy logic procedure in order to provide adequate bandwidth required to transmit the requests without missing their deadline as well as guarantee the fairness in the system; M_i is a fixed increment by one maximum packet length to guarantee the scheduler to at least transmit one packet in every round;

$$\gamma = \sum_{i=1}^N \gamma_i \quad (13)$$

Therefore, with regards to the fairness achievements, a queue with the lowest priority is allowed to transmit a minimum rate of $R_{min(i)}$ on every round and served prior to real-time applications when their deadlines have not been approached. In addition the amount of bits which remains in the existing round is kept in DC for the next transmission to avoid resource consumption and therefore maintain system performance.

Our aim is to find an optimal bandwidth γ for assignment to the service class queue. This is done by means of the fuzzy logic system that enables the scheduler at the BS to allocate fairly the bandwidth for the real-time flows within the delay bound.

With this method, our scheme calculates γ_i for each request in each service class queue, and then uses FADRR to manipulate the bandwidth granted to the queue. This also enables the non real-time in some cases to transmit ahead of real-time traffic when the deadline is observed. Here, linguistic forms in the fuzzy system are categorized by groups of linguistic relations. This relation forms a rule base of the fuzzy system which is transformed into a matrix equation (Driankov et al., 1996). This indicates that the behaviour of the output will vary depending upon the behaviour of the input parameters. Figure 2 describes the proposed embedded fuzzy system.

4.1.1. Fuzzy reasoning inference engine control model

In order to differentiate the QoS, WiMAX supports five service classes; all have their own queues Q_i ; $i=1:N$, where any application request with the same traffic type will be buffered in the corresponding Q_i .

Let Q_1 represents the queue for UGS service class, which requires highest priority due to the characteristics of this traffic type, and Q_N represent the queue for BE service class that has no delay and throughput requirements. The main goal of the embedded fuzzy system is to optimally specify the bandwidth for every service class queue in order to enable the scheduler to distinguish the serving priority at every round. The key reason to design fuzzy logic from fuzzy set theory is to design a theoretical structure for the linguistic information where the most important design of fuzzy logic based reasoning is to utilize the expert

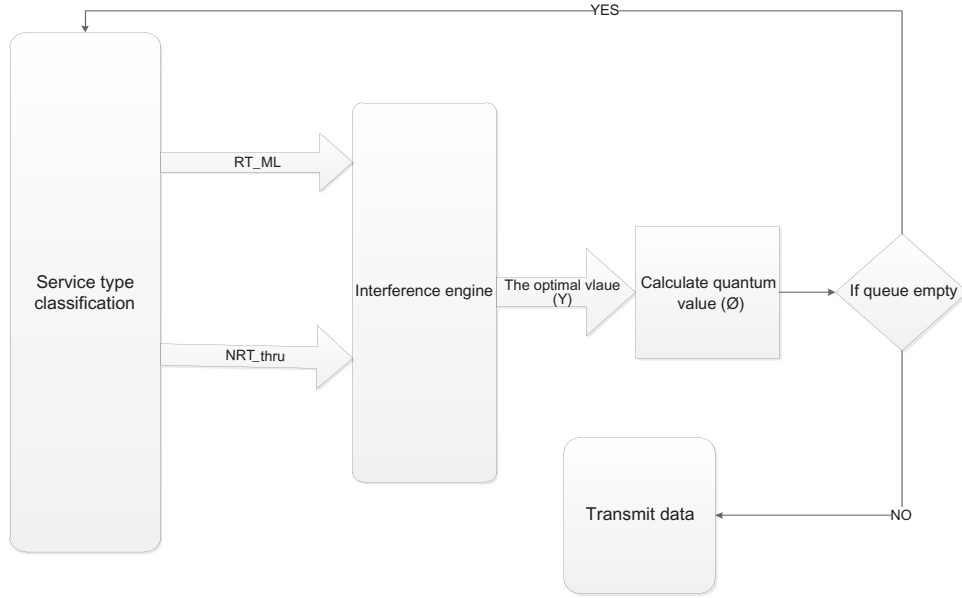


Fig. 2. Embedded fuzzy system proposed in the FADRR.

information for the rule base creation. In this system, we use individual based inference method with Mamdani's design (King and Mamdani, 1977), where the inference system rules are jointed into one value. The choice of this design was for flexibility of implementation.

This method consists of three fundamental mechanisms known as fuzzification, fuzzy reasoning inference and defuzzification. The role is to dynamically analyse all input traffic and combine them into one overall fuzzy set. Initially the fuzzification process handles two input variables, RT_{ML} and NRT_{Thru} for the overall system. Reasoning inference comes at the next stage which contains the rule base to manipulate the input variables as shown in Fig. 3. At this point, the actual decision is made representing the human expert process which performs to the linguistic behaviour to obtain the output value.

Lastly, the defuzzification phase calculates crisp numerical values to obtain the required weight, which provide indication of the priority for scheduler. Two expressions are classified into term sets that have a better response after numerous experiments. Hence, three terms for $T(RT_{ML})$ =(low, medium, high) and five terms for $T(NRT_{Thru})$ =(very low, low, medium, high, very high). Since, there are two input variables, the rule base becomes 15 with a dimension 3 for $|T(RT_{ML})| \times 5$ for $|T(NRT_{Thru})|$ as represented in Table 1. The dynamic scale normalized for the input and output variables are formed from 0 to 1 due to the change of the input variables namely maximum latency of real-time and throughput of non real-time packets for the input traffic.

4.2. Bandwidth assignment using FADRR

In order to utilize the required slots in the system to reach the highest overall system throughput taking into account the delay requirement for the service class traffic, FADRR considers real-time traffic with maximum latency, and non real-time traffic as the main factor to obtain the optimal bandwidth for UL sub-frame. In FADRR algorithm, there are several queues and each queue is attached with a DC where R_{min} are applied for each request. In every round, the DC is incremented by a ϕ value. The queue is transmitted when the DC is equal to the required amount of bandwidth. The fuzzy system determines ϕ value dynamically and employs intelligent strategy to allocate the right bandwidth to every queue in the system maintaining the overall system

capacity. The pseudo-code of FADRR is presented in Algorithms 1 and 2. FADRR algorithm maintains an active list of queues which consists of all the active flows with packets waiting in the queue. When a flow has no packet waiting in the queue, it is removed from the active list.

Algorithm 1. Pseudo-code of FADRR algorithm for uplink bandwidth allocation.

```

1: 1: Convert the bandwidth to physical slots Eq. (3)
2: 2: Verify the bandwidth requests at the BS
3: 3: Set initial  $\gamma_i = 0$ ,  $Active_{list} = 0$ ,  $DC = 0$ 
4: 4:  $\forall, Req_i \in Same\ ID\ in\ Active_{list}, do$ 
5: 5: Begin
6: 6: Calculate sum of all active queues Eq. (8)
7: 7: Assign  $w_i$  for all active queues Eq. (9)
8: for  $n = 1 : N$  do
9:   if  $bandwidth_{request(i)} = rtps$  then
10:    Calculate the deadline  $D = \frac{ML+SL_i}{FD}$ 
11:    if  $D_i < Second$  then
12:       $Uplink_{MAP} = grant_{bandwidth(i)}$ 
13:    else
14:       $\phi_i = \gamma \times \left( \frac{S_i \times \left( \frac{w_i}{D} \right)}{D} \right) + M_i$ 
15:       $grant_{bandwidth(i)} += grant_{bandwidth(i).slot[Size]}$ 
16:    end if
17:  end if
18:  if  $bandwidth_{request(i)} = nrtps$  then
19:     $w_i = \frac{R_{min(i)}}{\sum_{i=1}^N R_{min(i)}}$ 
20:     $grant_{bandwidth(i)} += bandwidth_{request(i).slot[Size]}$ 
21:    Assign the quantum value for the queue Eq. (12)
22:  end if
23:  if  $(Available_{Slot} > 0) \& (rtps, nrtps = empty)$  then
24:     $Uplink_{MAP} = grant_{bandwidth(BE)}$ 
25:  end if
26:  if  $DC > grant_{bandwidth(i)}$  then
27:    Move to the next not empty queue
28:  end if
29: end for
30: Return  $\leftarrow Uplink_{MAP}$ 

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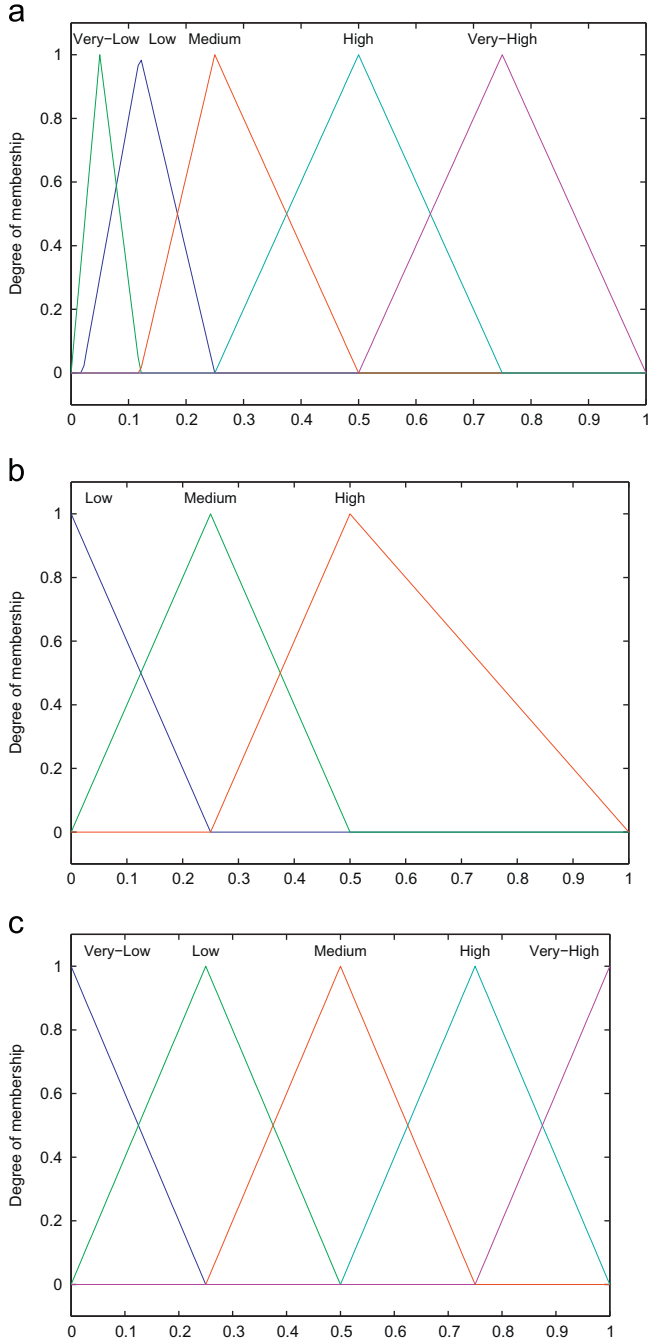


Fig. 3. Membership functions. (a) Throughput. (b) Maximum latency. (c) Output weight.

```

31:   Update the uplink MAP
32:   Allocate granted data to uplink sub-frame
33:   BS updates the active list status

```

Algorithm 2. Pseudo-code of the fuzzy inference algorithm.

```

1:   1: Read information from packets through parameters
2:   2: Check linguistic variable and member functions
3:   3: Compute the priority based on the rule base table
4:   if  $RT_{ML} = \text{Low}$  then
5:      $weight_{(r)} = \text{very high}$ 
6:   else
7:      $RT_{ML} = \text{medium}$ 
8:     if  $NRT_{Thru} = \text{very low, low OR medium}$  then

```

Table 1
Fuzzy rule base.

Rules no.	Throughput	Maximum latency	Output weight
1	Very low	Low	Very high
2	Low	Low	Very high
3	Medium	Low	Very high
4	High	Low	Very high
5	Very high	Low	Very high
6	Very low	Medium	High
7	Low	Medium	High
8	Medium	Medium	High
9	High	Medium	Medium
10	Very high	Medium	Medium
11	Very low	High	Medium
12	Low	High	Low
13	Medium	High	Low
14	High	High	Low
15	Very high	High	Very low

```

9:    $weight_{(r)} = \text{high}$ 
10:  else
11:    if  $NRT_{Thru} = \text{high OR very high}$  then
12:       $weight_{(r)} = \text{medium}$ 
13:    else
14:      if  $RT_{ML} = \text{high}$  then
15:        Check
16:      if  $NRT_{Thru} = \text{very low}$  then
17:         $weight_{(r)} = \text{medium}$ 
18:      else
19:        if  $NRT_{Thru} = \text{Low, medium OR high}$  then
20:           $weight_{(r)} = \text{Low}$ 
21:        else
22:          if  $NRT_{Thru} = \text{very high}$  then
23:             $weight_{(r)} = \text{very low}$ 
24:          while Number of Rules < 15 do
25:            Repeat step 1 till end of rule base table
26:          end while
27:        end if
28:      end if
29:    end if
30:  end if
31:  end if
32:  end if
33:  end if

```

5. Simulation model

This section describes a comparison between the proposed FADRR algorithm with MDRR (Cisco), and CDRR (Laia and Awan, 2010) using computer simulation. The system simulation is performed on a single BS in TDMA access mode using a point-to-multipoint (PMP) approach, whereby multiple SSs are uniformly distributed in 1.5 km radius as depicted in Fig. 4. Table 2 shows the parameters used in the system (Jain, 2008). The uplink sub-frame allocation is divided into a number of slots, and that all wireless connections between the BS and SSs are assumed to have complete information of the channel state.

The bandwidth allocation mechanism in this model deals with multimedia traffic. Specifically the flows of real-time and non real-time which are directed to their corresponding service classes queue UGS, ertPS, rtPS nrtPS, BE, respectively. The performance of the algorithm is evaluated in terms of delay, jitter, throughput and fairness. The traffic type, for instance VoIP for real-time application, measures the interval for the request polling every 20 ms.

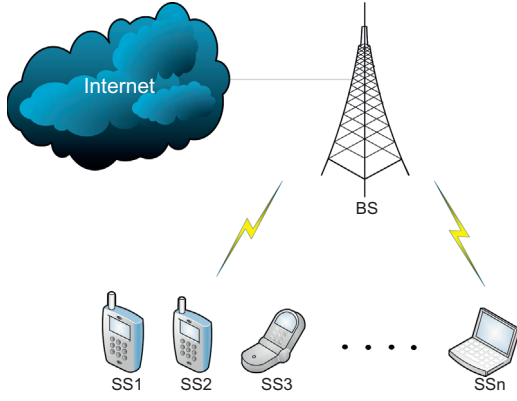


Fig. 4. Typical network topology.

Table 2
Simulation parameters.

Parameters	Value
Operating frequency	2.5 GHz
Channel bandwidth (MHz)	20
Gain (boresight)	16 dBi
Frame duration	5 ms
VoIP & Video Conference	< 150 ms
Latency	
FFT size	2048
Cell radius	1.5 km
Number of nodes	10–150
Number of BS	1
User distribution	Uniform
Mobility model	Random
Propagation model	Two ray
Beam pattern	Omni-directional

The applications contain video conference traffic representing the ertPS with traffic rate of 10 frames/s. nrtPS represents the demands of file transfer protocol (FTP) with traffic rate 512 kbytes where the polling interval service is achieved every 1 s. Finally BE represents HTTP with traffic rate (7247 bytes) where the BS allows multiple applications from a SS to transmit. Moreover, the system-level simulation is carried out for a number of SSs increased from 15 to 150 for every 10 units (Laia and Awan, 2010).

6. Results and discussion

In Fig. 5, we can see that the delay for FADRR is the smallest. This is because the real-time traffic must be handled with a certain latency constraint, where each request deadline is computed based on the deadline scheme derived from the embedded fuzzy logic system. This deadline calculation is based on the maximum latency for each request, which is manipulated as a weight factor and takes into account the scheduling delay to transmit the HoL packet in each queue. Therefore a high priority has to be given to real-time service types hence this service flow traffic enjoys the smallest end-to-end delay. In contrast, CDRR introduces a separate queue for those real-time requests whose times are going to be expired soon and hence it interrupts the scheduler to transmit their packets frequently. This increases delay in the overall system.

Figure 6 shows the average queuing delay for the three scheduling schemes. It can be seen again that FADRR reduces the packet queue delay because the traffic in the system is handled dynamically by the embedded fuzzy system method. It can be observed from the figure that as the number of SSs increases the

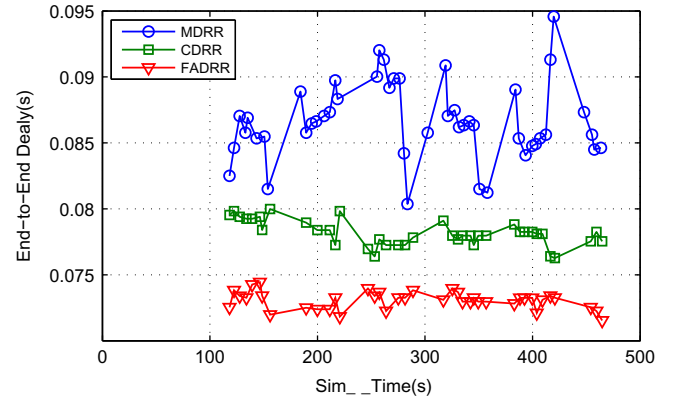


Fig. 5. The end-to-end delay for MDRR, CDRR and FADRR.

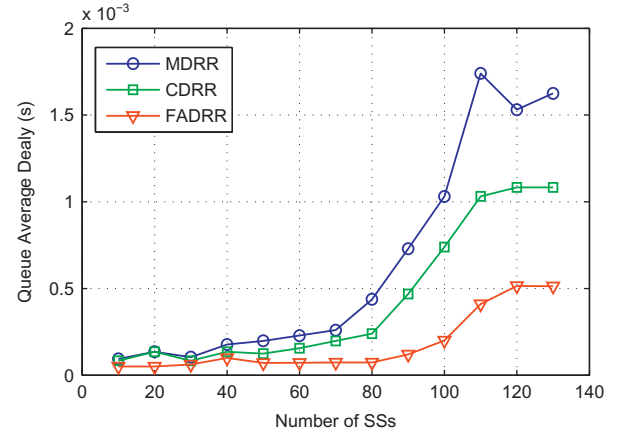


Fig. 6. The queue average delay for MDRR, CDRR and FADRR.

queuing delay in FADRR is maintained fairly constant, this is because the algorithm not only considers the HoL packet's time but is also aware of the scheduling time delay. On the other side, the average queue delay of MDRR and CDRR grows drastically at around 80 SSs; this arises because of the delay accumulated by the scheduling decision time performed at each round and because it does not consider the actual bandwidth required by each queue. Whereas in FADRR, at the point where SSs approach 120 the queue delay curve flattens out, which shows that the algorithm becomes stable because the different latency requirements have been met. This adequate bandwidth granted for each service class queue, enables each request to achieve their required data within the deadline constraints.

Fairness in scheduling algorithms is considered to have been achieved when the differences in the respective normalized received services flows are bounded (Laia and Awan, 2010; Lin et al., 2009). Here we evaluate the fairness based on the equation below in which the lowest values represent the most fair allocation.

$$\text{Fairness} = \left| \frac{\text{Thru}_{rtPS}}{S_{rtPS}} - \frac{\text{Thru}_{nrtPS}}{S_{nrtPS}} - \frac{\text{Thru}_{BE}}{S_{BE}} \right| \quad (14)$$

Where Thru_{rtPS} , Thru_{nrtPS} and Thru_{BE} represent the requested throughputs to the corresponding service class rtPS, nrtPS and BE, respectively; while S_{rtPS} , S_{nrtPS} and S_{BE} represent the corresponding bandwidths, respectively.

Fairness is very significant since connected SSs expect to have the same response, regardless either they are requesting for real-time or non real-time traffic. Figure 7 shows a comparison of the fairness for FADRR with MDRR and CDRR. It can be seen that MDRR fairness deteriorates as traffic load increases, because it

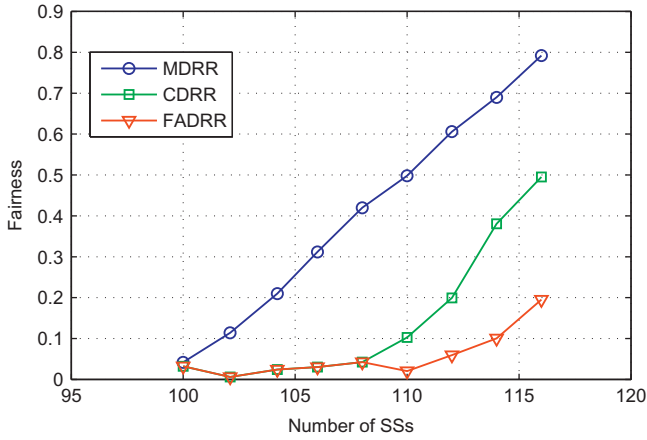


Fig. 7. Fairness of MDRR, CDRR and FADRR.

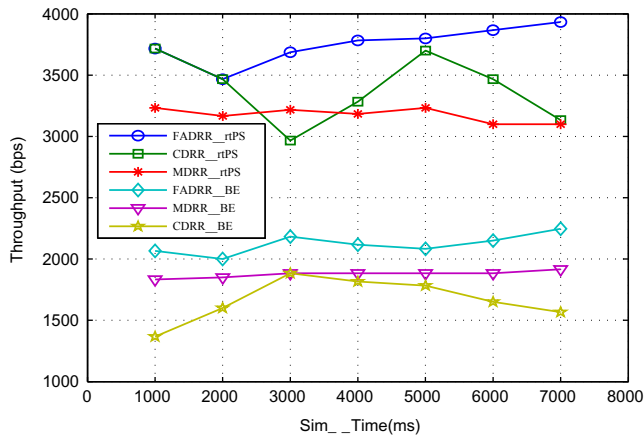


Fig. 8. Throughput of rtPS and BE traffic for MDRR, CDRR and FADRR.

gives high priority for real-time application. Consequently, non real-time tends to get starved when real-time SSs connected. On the other side CDRR shows stable in fairness when the system load below 110 SSs, because it employs extra queue for real-time applications. However, these extra queues are starved the bandwidth for non real-time application when the system loaded exceeds 110 SSs. This is due to the non real-time applications engaged for a long time, whereas the DC has not yet finished serving real-time service type queue. In contrast, FADRR shows better fairness than MDRR and CDRR even when 120 SSs are involved. The degraded fairness observed in FADRR because of real-time applications approaching the deadline are given higher priority. However, FADRR still maintains non real-time traffic within their minimum reserved rate which gives a better fairness against MDRR and CDRR.

Figure 8 compares the throughput for BE and rtPS for three scheduling algorithms. We focus on these two types of services in which BE is of a lesser priority type and rtPS represents real-time traffic. It is clear that the throughput for BE traffic is generally lower than that of rtPS, because priority in the scheduler is given to real-time traffics. The throughput of BE traffic in FADRR compared with MDRR and CDRR, is maintained at a certain minimum reserved rate which is still higher than MDRR or CDRR. This increased throughput is observed over a certain time period. This stems from the fact that, since FADRR gives adequate bandwidth to rtPS queues, this enables the scheduler to serve the BE traffic first, when rtPS still has sufficient time to spare before its deadline expires. However, throughput noticeably is smaller for BE compared to rtPS, this is because the algorithm allocates much more bandwidth for rtPS flows than for BE.

In the other hand CDRR throughput noticeably decreases for BE, this is because the extra queue implemented for rtPS flows. Eventually starve the BE flows and degrades the throughput of the system.

Figure 9 shows the average delay of the various service classes achieved by FADRR. It could be seen that the delay of UGS service class is bounded; this is because the BS scheduler gives higher priority to UGS besides giving it fixed bandwidth. In contrast, the BE and nrtPS packets experience increased delay when the number of SSs increases; this is because more slots are granted to satisfy the real-time traffic. However, this does not indicate that the BE flows will be starved because FADRR guarantee the minimum reserved rate for non real-time applications. On the other hand, ertPS and rtPS packets only experience a lower delay, since FADRR can maintain the maximum latency and distinguish the allocation priority for their corresponding service requirements.

Figure 10 shows the throughput of the various service classes in FADRR; we can see that the throughput is increases as a function of the number of SSs and the service classes. The high level of throughput for UGS and ertPS service classes is attributed to the increased demands of service type in the evaluated system as well as the fixed amount of bandwidth granted by the system. On the other hand, we can see that the throughput of rtPS, nrtPS and BE lie close to each other as small number of SSs. This is due to adequate bandwidth allocation. As a lowest priority nrtPS and BE their higher throughput is relevant to rtPS traffic when the number of SSs reached 60. This is because FADRR gives a higher priority to nrtPS and BE than rtPS when their packets do not

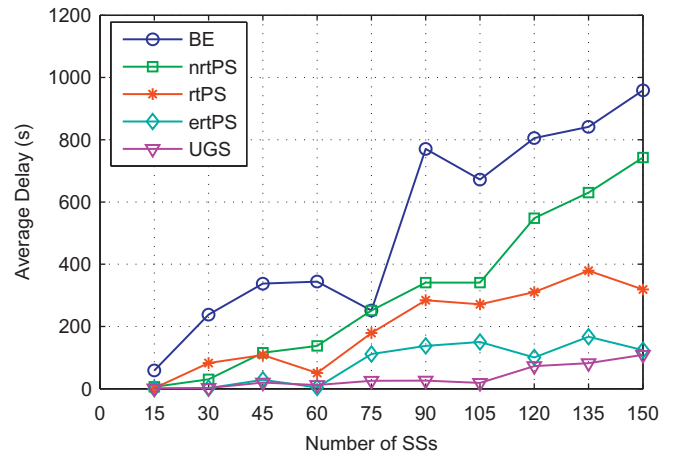


Fig. 9. The average delay for FADRR in service classes.

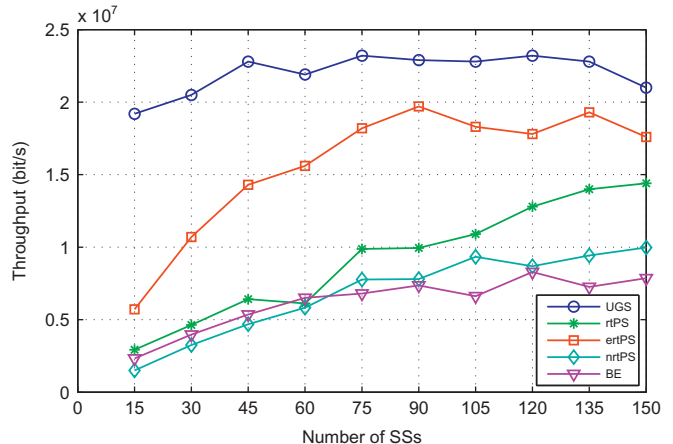


Fig. 10. Throughput for FADRR for the various service classes.

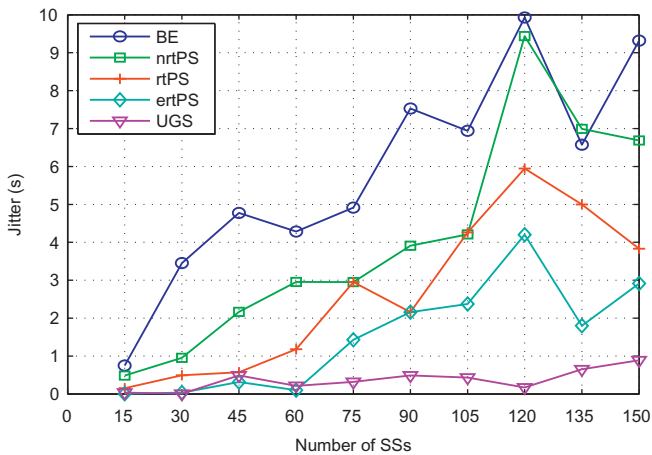


Fig. 11. Delay jitters of all service classes for FADRR.

violate the maximum latency bound. This enhances the system throughput without sacrificing QoS requirements.

Figure 11 shows the average jitter of the various service classes for FADRR, jitter is defined as variations in delay of the packets arriving at the destination. We can see that the average jitter of UGS is smallest and increases slightly when the number of SSs approaches 150. This is because the queue at this point (SS=150) is expected to be full due to the increased traffic so further incoming packets to the same queue will be discarded. However, this jitter is still considered very small compared with the heavy flow handled by the system. In respect to the remaining service classes, we can see that due to the network delay and the higher amount of slots allocated to corresponding applications, the jitter in these service classes as expected is significantly affected. Jitter in ertPS and rtPS is moderate in respect to the increased number of SSs, and this is consider reasonable in view of the heavy flow handled by the system. However, the number of slots allocated for all service classes is adequate as FADRR is aware of the delay bound and the overall system throughput through the embedded intelligent system in the scheduler to satisfy the QoS requirements. Finally, it is observed that the jitter of nrtPS and BE increase tremendously for high numbers of SSs. This is expected because of the associated increase in traffic and its lower weightage in the queue. However, this is not critical for the service, since BE and nrtPS are not sensitive to jitter.

7. Conclusion

In this paper, a new bandwidth allocation algorithm called FADRR for the uplink transmission in mobile WiMAX network has been described. FADRR is fully dynamic, using Fuzzy logic based approach and adaptive of the various service type flows in the BS. FADRR presents a new adaptive deadline-based approach in order to allocate, dynamically and optimally, bandwidth for real-time and non real-time applications. The optimal bandwidth allocated for each service type is derived by means of a fuzzy expert system that grants the optimal bandwidth required by each flow, based on maximum latency and throughput. It also considers the deadline and required bandwidth for each request received by the BS.

The overall system throughput in FADRR is stabilized by improving the average fairness in the system. This is achieved by giving non real-time traffic transmission priority over that of real-time traffic when the deadline of the real-time traffic can still be

tolerated. FADRR has also been evaluated against MDRR and CDRR scheduling schemes in terms of delay, jitter, throughput and fairness for different service classes namely UGS, ertPS, rtPS, nrtPS and BE. Simulation results show that our FADRR scheduling algorithm is efficient in respect of QoS parameter for real-time applications, while it also gives fair allocation to non real-time applications and optimize the overall system throughput.

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